

Lab: Production-Grade Highly Available Web Application on EC2

Objective: Build a complete, cost-optimized, resilient 3-tier web application using EC2 best practices.

Skills Covered:

- EC2 Launch, AMIs, Instance Types
- Auto Scaling + Load Balancing.
- EBS Volumes, Snapshots, Encryption

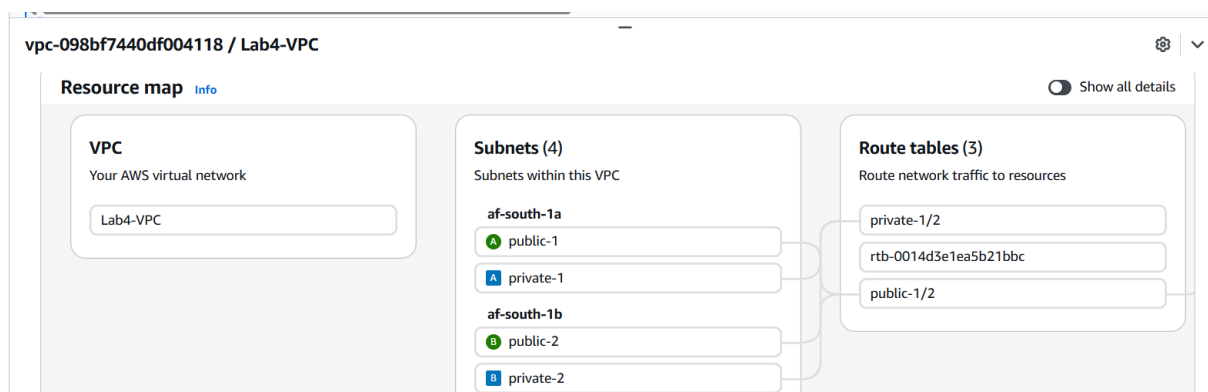
Region: af-south-1

Architecture Overview

- **ALB (Public)** → Distributes traffic
- **Auto Scaling Group (Private Subnets)** → 1-6 EC2 instances
- **EBS gp3 Volumes** with snapshots + encryption
- **Mixed Pricing:** On-Demand + Savings Plans awareness

Phase 1: Infrastructure Setup

- Create a simple **VPC**:
 - 2 Public Subnets + 2 Private Subnets



- Create a **Launch Template** (not Launch Configuration):
 - AMI: **Amazon Linux 2023**
 - Instance Type: t3.micro (good balance)
 - Security Group: Allow HTTP/HTTPS from ALB + SSH from your IP
 - User Data (demo page):

```
#!/bin/bash
```

```
# Week 4 Capstone - User Data Script
```

```
# Update system
```

```
yum update -y
```

```
# Install Apache
```

```
yum install -y httpd
```

```
# Start and enable Apache
```

```
systemctl start httpd
```

```
systemctl enable httpd
```

```
# Create a nice demo page
```

```
cat > /var/www/html/index.html << EOF
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Week 4 Capstone</title>
```

```
<style>
```

```
body { font-family: Arial; text-align: center; margin-top: 50px; }
```

```
h1 { color: #232F3E; }
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1> Week 4 Capstone Lab</h1>
```

```
<h2>Served from Instance: $(hostname -f)</h2>
```

```

<p>Auto Scaling Group + ALB is working!</p>

<p>Instance ID: $(curl -s
http://169.254.169.254/latest/meta-data/instance-id)</p>

</body>

</html>

EOF

# Restart Apache to load the new page

systemctl restart httpd

echo "User Data script executed successfully at $(date)" >>
/var/log/user-data.log

```

Launch Templates (1/1) [Info](#)

☒	Launch Template ID	Launch Template Name	Default Version	Latest Version	C
☒	lt-0cb3ae3b7c007429f	Lab-4	1	1	2

Lab-4 (lt-0cb3ae3b7c007429f)

Launch template version details

Version
1 (Default)

Description
lab4

Date created
2026-05-21T13:14:33.000Z

Instance details

Storage

Resource tags

Network interfaces

Advanced details

AMI ID
ami-0ace8223c93536b12

Instance type
t3.micro

Availability Zone
-

Key pair name
Lab4

Security groups
-

Security group IDs
sg-094f8cf6135b9e0f3

Phase 2: Storage & Security

- **Create an Encrypted EBS Volume:**
 - gp3, 10 GB
 - Enable encryption (default KMS key)

EC2 > Volumes > vol-00690eab41bb3b7ff

AMI Catalog

- ▼ Elastic Block Store
 - Volumes
 - Snapshots
 - Lifecycle Manager
- ▼ Network & Security
 - Security Groups
 - Elastic IPs
 - Placement Groups
 - Key Pairs
 - Network Interfaces
- ▼ Load Balancing
 - Load Balancers
 - Target Groups
 - Trust Stores
- ▼ Auto Scaling
 - Launch Configurations

vol-00690eab41bb3b7ff

Last updated less than a minute ago [Actions](#) [Delete](#) [Modify](#)

Details			
Volume ID vol-00690eab41bb3b7ff	Size 10 GiB	Type gp3	Status check ✓ Okay
AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more	Volume state ✓ Available	IOPS 3000	Throughput 125
Fast snapshot restored No	Availability Zone afs1-az1 (af-south-1a)	Created Fri May 22 2026 18:44:32 GMT+0100 (West Africa Time)	Multi-Attach enabled No
Attached resources -	Outposts ARN -	Managed false	Operator -
▼ Source			
Snapshot ID -	Source volume ID -		
▼ Encryption			

- Launch one manual EC2 first in a private subnet using the Launch Template.

Instances (1/1) Info

Saved filter sets [Choose filter set](#) [Refresh](#) [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

☒	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☒		i-067461d25cf27957e	Running	t3.micro	3/3 checks passed	af-south-1a	-

- Attach the encrypted volume

Successfully attached volume vol-00690eab41bb3b7ff to instance i-067461d25cf27957e.

Volumes (1/2) Info

Saved filter sets [Choose filter set](#) [Refresh](#) [Recycle Bin](#) [Actions](#) [Create volume](#)

☐	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume ID	Created
☐		vol-073fcf727d745cab8	gp3	8 GiB	3000	125	snap-0bcbfd...	-	2026/05/22 18:44:32
☒		vol-00690eab41bb3b7ff	gp3	10 GiB	3000	125	-	-	2026/05/22 18:44:32

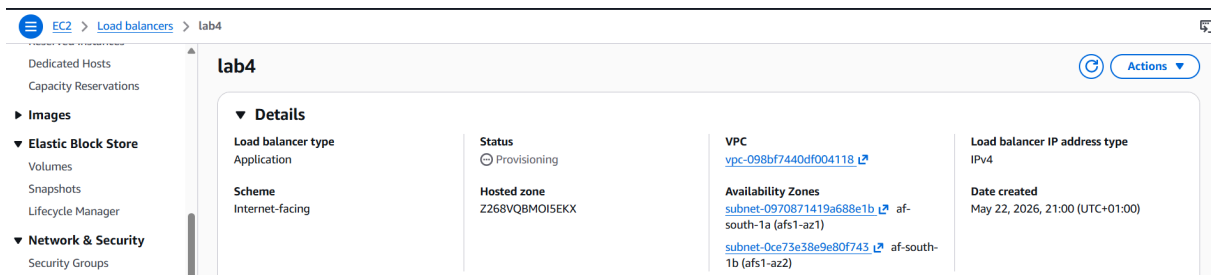
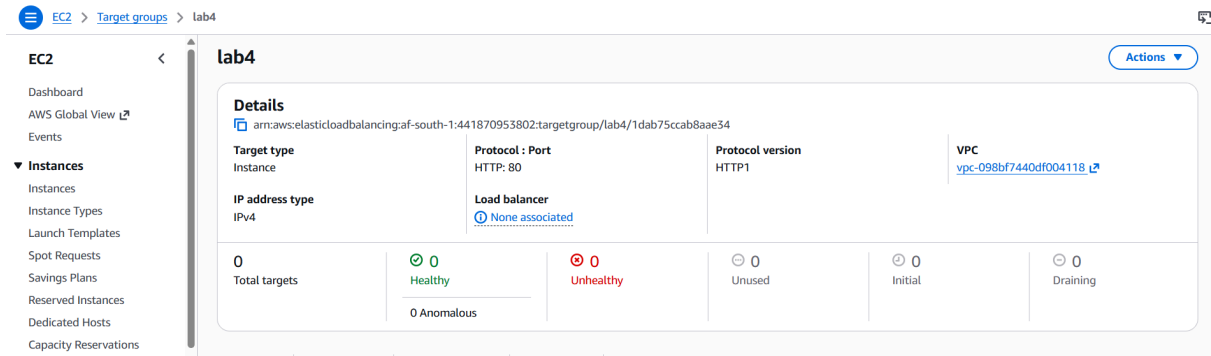
Volume ID: vol-00690eab41bb3b7ff

Details	Status checks	Monitoring	Tags
Volume ID vol-00690eab41bb3b7ff	Size 10 GiB	Type gp3	Status check ✓ Okay
AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more	Volume state ✓ In-use	IOPS 3000	Throughput 125
Fast snapshot restored No	Availability Zone afs1-az1 (af-south-1a)	Created Fri May 22 2026 18:44:32 GMT+0100 (West Africa Time)	Multi-Attach enabled No
Attached resources i-067461d25cf27957e (EBS card index: 0): /dev/sdf (attached)	Outposts ARN -	Managed false	Operator -

Phase 3: Auto Scaling + Load Balancing

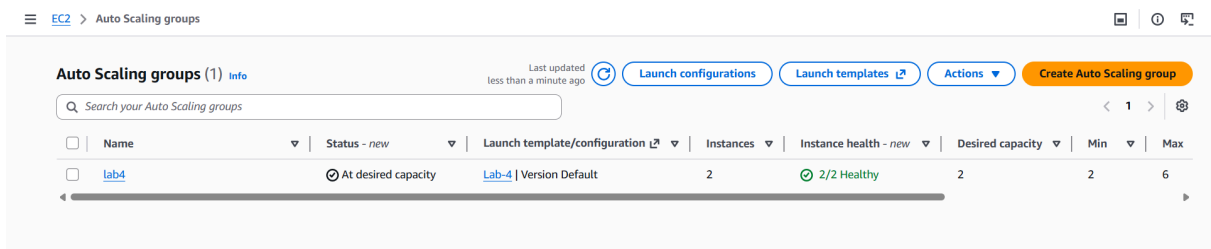
- **Create an Application Load Balancer (ALB):**

- Internet-facing
- Target group pointing to your ASG



- **Create Auto Scaling Group:**

- Use the Launch Template
- Min: 1, Desired: 2, Max: 6
- Target Tracking Policy: Average CPU = 50%
- Subnets: Your two public subnets
- Attach to the ALB target group



- **Test scaling:**

- Stress the instances (use stress tool or just open many browser tabs)
- Watch new instances launch automatically



Week 4 Capstone Lab

Served from Instance: ip-10-0-0-37.af-south-1.compute.internal

Auto Scaling Group + ALB is working!

Instance ID:



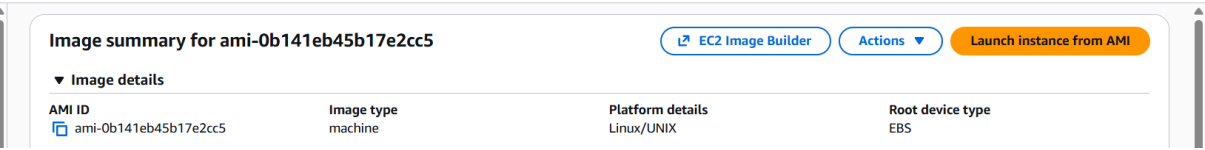
Week 4 Capstone Lab

Served from Instance: ip-10-0-0-125.af-south-1.compute.internal

Auto Scaling Group + ALB is working!

Instance ID:

9. Create an AMI from your configured EC2 instance (for future use).



Thank You